

ENEL700 Tutorial 2

1. What is the gain of an amplifier with an output of 1.5V and an input of $30\mu\text{V}$?
2. A piece of communication equipment has two stages of amplification with gains of 40 and 60 and two loss stages with attenuation factors of 0.03 and 0.075. The output voltage is 2.2V. What are the overall gain (or attenuation) and the input voltage?
3. A power amplifier has a gain of 55dB. The input power is 600mW. What is the output power?
4. What is the value of inductance that will resonate with a 80pF capacitor at 18MHz?
5. Sketch the amplitude spectra of the following signal –
$$v(t) = 5 + 50\sin(4 \times 10^6 t + 45^\circ) - 45\sin(8 \times 10^6 t) + 30\sin(1.2 \times 10^5 t - 60^\circ)$$

and determine the average power contained in the waveform.